

Basics of Polynomials

• Exercise 1

Let $p(x) = x^3 + 2x^2 - x + 1$ and $q(x) = -x^3 + x^2 + 3x - 2$.

1. Compute $(p + q)(x)$ and $(pq)(x)$.
2. Find $\deg p$, $\deg q$, $\deg(p + q)$, and $\deg(pq)$.

◦ Solution 1.1

1. For the sum of the polynomials, we have:

$$\begin{aligned}(p + q)(x) &= p(x) + q(x) \\ &= (x^3 + 2x^2 - x + 1) + (-x^3 + x^2 + 3x - 2) \\ &= 3x^2 + 2x - 1.\end{aligned}$$

For the product of the polynomials, by expanding each term systematically, we have:

$$\begin{aligned}(pq)(x) &= p(x)q(x) \\ &= (x^3 + 2x^2 - x + 1)(-x^3 + x^2 + 3x - 2) \\ &= x^3(-x^3 + x^2 + 3x - 2) + 2x^2(-x^3 + x^2 + 3x - 2) \\ &\quad - x(-x^3 + x^2 + 3x - 2) + 1(-x^3 + x^2 + 3x - 2) \\ &= (-x^6 + x^5 + 3x^4 - 2x^3) + (-2x^5 + 2x^4 + 6x^3 - 4x^2) \\ &\quad + (x^4 - x^3 - 3x^2 + 2x) + (-x^3 + x^2 + 3x - 2) \\ &= -x^6 + (1 - 2)x^5 + (3 + 2 + 1)x^4 + (-2 + 6 - 1 - 1)x^3 \\ &\quad + (-4 - 3 + 1)x^2 + (2 + 3)x - 2 \\ &= -x^6 - x^5 + 6x^4 + 2x^3 - 6x^2 + 5x - 2.\end{aligned}$$

2. By definition, we obtain that

- Since the leading term of $p(x)$ is x^3 , we have $\deg p = 3$.
- Since the leading term of $q(x)$ is $-x^3$, we have $\deg q = 3$.
- Since the leading term of $(p + q)(x) = 3x^2 + 2x - 1$ is $3x^2$, so we have $\deg(p + q) = 2$.
- Since the leading term of $(pq)(x) = -x^6 - x^5 + 6x^4 + 2x^3 - 6x^2 + 5x - 2$ is $-x^6$, so we have $\deg(pq) = 6$.

• Exercise 2

Given $n \in \mathbb{N}$ and $a \in \mathbb{K}$, perform the following products of polynomials.

1. $(x - 1)(x^{n-1} + ax^{n-2} + \dots + a^{n-2}x + a^{n-1})$.
2. $(x + 1)(x^2 + 1)(x^4 + 1) \dots (x^{2^n} + 1)$.

◦ Solution 2.1

$$\begin{aligned}1. \quad (x - a)(x^{n-1} + ax^{n-2} + \dots + a^{n-2}x + a^{n-1}) &= (x^n + ax^{n-1} + a^2x^{n-2} + \dots + a^{n-1}x) \\ &\quad - (ax^{n-1} + a^2x^{n-2} + \dots + a^{n-1}x + a^n) \\ &= x^n - a^n.\end{aligned}$$

2. We consider the product based on the value of x :

If $x = 1$, then

$$\begin{aligned}(1 + 1)(1^2 + 1)(1^4 + 1) \dots (1^{2^n} + 1) &= \underbrace{2 \cdot 2 \cdot 2 \dots 2}_{n+1 \text{ times}} \\ &= 2^{n+1}.\end{aligned}$$

If $x \neq 1$, then

$$(1+1)(1^2+1)(1^4+1)\cdots(1^{2^n}+1) = \underbrace{2 \cdot 2 \cdot 2 \cdots 2}_{n+1 \text{ times}} = 2^{n+1}.$$

If $x \neq 1$, then

$$\begin{aligned} (x+1)(x^2+1)(x^4+1)\cdots(x^{2^n}+1) &= \frac{(x-1)(x+1)(x^2+1)(x^4+1)\cdots(x^{2^n}+1)}{x-1} \\ &= \frac{(x^2-1)(x^2+1)(x^4+1)\cdots(x^{2^n}+1)}{x-1} \\ &= \frac{(x^4-1)(x^4+1)\cdots(x^{2^n}+1)}{x-1} \\ &\vdots \\ &= \frac{(x^{2^n}-1)(x^{2^n}+1)}{x-1} \\ &= \frac{x^{2^{n+1}}-1}{x-1} \\ &= x^{2^{n+1}-1} + x^{2^{n+1}-2} + \cdots + x + 1. \end{aligned}$$

Evaluating this resulting geometric series at $x = 1$ yields $\sum_{k=0}^{2^{n+1}-1} 1^k = 2^{n+1}$, which matches the first case. Therefore, for all cases, we obtain that

$$(x+1)(x^2+1)(x^4+1)\cdots(x^{2^n}+1) = x^{2^{n+1}-1} + x^{2^{n+1}-2} + \cdots + x + 1.$$

• **Exercise 3**

Find the quotient and the remainder of the division of $3x^3 - 16x^2 + 23x - 2$ by $3x - 1$.

◦ **Solution 3.1**

We have that $3x^3 - 16x^2 + 23x - 2 = (3x - 1)(x^2 - 5x + 6) + 4$. Thus $x^2 - 5x + 6$ is the quotient and 4 is the remainder.

• **Exercise 4**

Let $a, b \in S, a \neq 0$. For $f \in S[x] \setminus \{0\}$, prove that the remainder of the division of f by $ax + b$ is $f\left(-\frac{b}{a}\right)$.

◦ **Proof 4.1**

By the division algorithm for polynomials, there exist unique polynomials $q(x), r(x) \in S[x]$ such that

$$f(x) = (ax + b)q(x) + r(x)$$

with either $r(x) = 0$ or $\deg(r(x)) < \deg(ax + b) = 1$. This implies that $r(x)$ must be a constant polynomial, so we can write $r(x) = c$ for some $c \in S$.

We now evaluate the polynomial equation $f(x) = (ax + b)q(x) + c$ at the point $x = -\frac{b}{a}$. This yields the following expression

$$f\left(-\frac{b}{a}\right) = \left(a\left(-\frac{b}{a}\right) + b\right)q\left(-\frac{b}{a}\right) + c = (-b + b)q\left(-\frac{b}{a}\right) + c = 0 \cdot q\left(-\frac{b}{a}\right) + c = c.$$

Since the constant remainder c is exactly equal to $f\left(-\frac{b}{a}\right)$, substituting this back into the division equation proves that the remainder is $f\left(-\frac{b}{a}\right)$.

- **Exercise 5**

For which values of a and b does the polynomial $p(x) = 3x^3 + ax^2 + bx + 2$ have a remainder equal to 12 when divided by $x + 1$ and a remainder of 4 when divided by $x + 2$?

- **Solution 5.1**

By the Remainder Theorem, the remainder of the division of $p(x)$ by $x - \alpha$ is equal to $p(\alpha)$.

For the divisor $x + 1 = x - (-1)$, the remainder is $p(-1) = 12$. Substituting $x = -1$ into $p(x)$ gives the first equation

$$3(-1)^3 + a(-1)^2 + b(-1) + 2 = 12 \implies -3 + a - b + 2 = 12 \implies a - b = 13. \quad (1)$$

For the divisor $x + 2 = x - (-2)$, the remainder is $p(-2) = 4$. Substituting $x = -2$ into $p(x)$ gives the second equation

$$3(-2)^3 + a(-2)^2 + b(-2) + 2 = 4 \implies 2a - b = 13. \quad (2)$$

We now solve the linear system of equations formed by (1) and (2). Subtracting equation (1) from equation (2) eliminates b and yields

$$(2a - b) - (a - b) = 13 - 13 \implies a = 0.$$

Substituting $a = 0$ back into equation (1) gives

$$0 - b = 13 \implies b = -13.$$

Therefore, the required values are $a = 0$ and $b = -13$.